

A general nonparametric framework for testing hypotheses about function-valued parameters

Albert Osom ^{*1}, Ali Shojaie^{1,2}, and Aaron Hudson ^{1,3}

¹Department of Biostatistics, University of Washington

²Department of Statistics, University of Washington

³Vaccine and Infectious Disease Division, Fred Hutchinson Cancer Center

Abstract

We present a general nonparametric approach for testing whether a statistical parameter defined through conditional distributions is constant across covariates. Such hypotheses arise naturally in problems such as assessing treatment effect heterogeneity, conditional associational effects, and conditional mean dependence. Our framework studies function-valued parameters obtained by evaluating finite-dimensional statistical functionals on conditional probability distributions. We establish an explicit connection between our test and procedures based on studying the norm of the function-valued parameter. Unlike many existing norm-based tests, which exhibit poor asymptotic behavior under the null, the proposed test statistic admits a tractable limiting null distribution. We illustrate the applicability of the proposed test through several examples, assess its operating characteristics in simulation studies, and apply it to data from a breast cancer trial to identify predictive biomarkers for response to adjuvant chemotherapy.

1 Introduction

Testing hypotheses about smooth function-valued parameters, such as conditional mean functions, and conditional average treatment effect (CATE) curves, is a commonly-occurring problem in many scientific applications, and it presents a fundamental challenge, particularly in nonparametric models. For instance, clinical investigators often seek to determine whether the effect of an intervention on

^{*}Corresponding author: aosom@uw.edu

a health outcome differs across subgroups defined by patient characteristics, or if certain patient characteristics predict a health outcome. Addressing these questions may require performing inference on CATE or a conditional mean function. Both of these parameters, which we will study later, fall within a broader class of function-valued parameters that can be expressed as an evaluation of a finite-dimensional statistical functional on a conditional probability distribution. The current paper focuses on this class of function-valued parameters, which encompasses many other parameters, including the conditional covariance function, the conditional survival function, and linear transformations of the conditional mean function (Chernozhukov et al., 2024). We develop a general framework for testing the null hypothesis that the evaluation of a given functional on a conditional probability distribution is constant in the conditioning variable. For example, this null can correspond with equality of marginal and conditional means in a mean regression problem.

1.1 Related literature

Inference on function-valued parameters has been a topic of interest for many years, but methodological work remains limited, especially when aiming to remain agnostic to the data-generating mechanism. In contrast, if one restricts the space of distributions that generate the data (e.g., parametric models) or the structure of the function-valued parameters (e.g., linearity), inference becomes much simpler and has been well studied. For instance, to test whether a variable is a significant predictor of the outcome in a linear regression model, it suffices to test whether its coefficient in the model is zero. Although this approach is straightforward, we risk deriving misleading conclusions if the model assumptions are incorrect or misspecified (Whitney et al., 2020). To mitigate this limitation, it is natural to relax the restrictions on the model space and allow the function-valued parameter to fall within a nonparametric or semi-parametric model. Inference under this relaxed model space is challenging, as it often requires flexibly estimating the function-valued parameter using machine learning or data-adaptive approaches, which typically do not converge at the standard parametric $n^{1/2}$ convergence rate. Despite these challenges, general inferential strategies have been proposed to perform inference on certain classes of function-valued parameters under nonparametric models (Luedtke et al., 2019; Hudson, 2023; Westling, 2022).

Existing hypothesis tests for function-valued parameters, including those proposed in Luedtke et al. (2019), Hudson (2023), and Westling (2022), rely on test statistics that are based on a one-

dimensional summary of the function-valued parameter. In particular, [Westling \(2022\)](#) introduce a test statistic based on an estimator of the primitive (i.e., integrated) form of the causal dose-response function to assess whether it is constant. Their approach can be used to test hypotheses about other function-valued parameters (see, e.g., [Wang et al., 2024](#)), but the test may suffer from low power in finite samples when the function-valued parameter is highly complex or non-monotone. [Hudson et al. \(2023\)](#) and [Jin et al. \(2024\)](#) develop generalized tests for constancy of the dose-response function that can adapt to non-monotonicity. [Hudson \(2023\)](#) focus on inference on function-valued parameters that can be expressed as the minimizer of a goodness-of-fit measure (e.g., expected squared error). They construct their test statistic based on the maximum difference between the estimator of the minimum goodness-of-fit measure and the goodness-of-fit measure evaluated at the null parameter along a path that passes through the function-valued parameter. However, if the function-valued parameter of interest is not a minimizer of some goodness-of-fit measure, their approach is inapplicable. [Luedtke et al. \(2019\)](#) propose a test statistic based on the mean discrepancy ([Gretton et al., 2006](#)) between the distributions of the function-valued parameter and the null parameter, maximized within the unit reproducing kernel Hilbert space (RKHS). Although this framework is more broadly applicable, it requires second-order pathwise differentiability ([Robins et al., 2008, 2009](#); [van der Vaart, 2014](#); [Carone et al., 2014](#)) to avoid degeneracy issues in characterizing the limiting null distribution of the test statistic. Even when second-order pathwise differentiability holds, obtaining second-order influence functions remains challenging ([Carone et al., 2014](#); [Fisher and Kennedy, 2021](#)).

1.2 Our contributions

We propose a nonparametric test for a class of null hypotheses about the constancy of a function-valued parameter obtained as evaluation of a finite-dimensional statistical parameter at a conditional distribution. We develop a norm-based test that leverages the representation of norms as a supremum taken over a collection of finite-dimensional statistical parameters indexed by functions in a prescribed class. These finite-dimensional parameters are estimated using debiased machine-learning techniques ([Chernozhukov et al., 2018a](#)) that attain $n^{1/2}$ -rate of convergence without requiring sample splitting, and their supremum serves as the test statistic. We show that our test statistic has a well-behaved limiting null distribution and admits an asymptotic representation based on influence functions. Moreover, using the the supremum representation of the norm allows us to choose function classes that

leverage structural assumptions on the function-valued parameter for more efficient estimation of the norm. Our test is guaranteed to achieve asymptotic type-I error control, and its power against a given alternative depends on the choice of the function class. To ensure power against a wide range of alternatives, we propose an aggregate test that combines several (user-specified) function classes. Additionally, we describe a broad class of function-valued parameters for which our proposed test statistic can be constructed without explicitly deriving influence functions from scratch. This feature makes the framework easy to specialize to additional examples without requiring as much specialized knowledge about semi-/non-parametric efficiency theory.

Our proposed testing procedure has the following four advantages over existing methods. Firstly, our test can be more powerful than tests based on the norm of the function-valued parameter. To establish this, we explicitly connect our test to tests based on the norm of the function-valued parameter. These tests are valid since the norm is zero if and only if the null is true. However, the sample estimates of the norm does not admit to a tractable limiting distribution under the null ([Williamson et al., 2023](#); [Dai et al., 2022](#)). Though one can obtain well-behaved norm-estimators using sample-splitting ([Williamson et al., 2023](#); [Dai et al., 2022](#); [Cai et al., 2024](#)), this approach results in low power due to inefficient use of the data. Our test avoids these issues and establishes a theoretical size guarantee, without the need for sample-splitting. Secondly, our framework allows seamless construction of test statistics that can leverage known or assumed structure on the function-valued parameter—e.g., monotonicity, sparsity, and smoothness—to improve power, using a similar strategy as described in [Jin et al. \(2024\)](#). Thirdly, we introduce an aggregate test that combines different individual test statistics, while preserving tight size control. Lastly, our proposed test is easy to implement and readily generalizes to new testing problems. Unlike the method of [Luedtke et al. \(2019\)](#), our method requires only first-order pathwise derivatives, and we provide theoretical results that greatly simplify these derivations, eliminating the need for manual calculations in many settings. Our test also applies to classes of function-valued parameters that are not covered by the method of [Hudson et al. \(2023\)](#) and are not addressed in [Luedtke et al. \(2019\)](#).

The remainder of the paper is organized as follows. Section 2 presents the general testing problem and introduces two main illustrative examples: testing conditional mean dependence and testing treatment effect heterogeneity. Section 3 describes our testing strategy and the proposed test statistic. We apply our framework to the illustrative examples in Section 4. In Section 5, we examine some

choices of function classes over which the supremum can be evaluated and describe an aggregate test for combining multiple test statistics. Section 6 evaluates the performance of our method through simulation studies, and Section 7 presents a data example. Section 8 provides concluding remarks and directions for future research.

2 Overview of the testing problem and examples

Let $O = (U, V)$ be a random vector drawn from a distribution P_0 with support $\mathcal{O} = \mathcal{U} \times \mathcal{V}$, where $\mathcal{U}$ and $\mathcal{V}$ are the supports of U and V , respectively. Suppose we observe $\{O_i = (U_i, V_i)\}_{i=1}^n$ independent and identically distributed (i.i.d.) copies of $O = (U, V)$ from the distribution P_0 . We let $P_0 \in \mathcal{M}$, where $\mathcal{M}$ denotes the collection of all distributions on $\mathcal{O}$. Let $P_{0,U}$, $P_{0,V}$, $P_{0,U|V}$, and $P_{0,O|V}$ denote the respective marginal and conditional distributions derived from P_0 . We use $o = (u, v)$ to denote a realization of the random vector $O = (U, V)$. For any $P \in \mathcal{M}$, let $\Psi(P)$ denote a smooth functional. We assume that Ψ is pathwise differentiable (Bickel et al., 1993) at P_0 . Our goal is to make inference on the corresponding function-valued parameter obtained by applying the same functional to the conditional distribution of O given V . For notational simplicity, we write $\Psi_0 = \Psi(P_0)$ and $\Psi_{0,V} = \Psi(P_{0,O|V})$. For example, $\Psi_0 = \int u dP_0(o)$ and $\Psi_{0,V} = \int u dP_{0,O|V}(o)$. We define the class of P_0 p -integrable functions with induced norm $\|f\|_{L^p(P_0)}$ as $L^p(P_0) := \{f : \mathcal{O} \rightarrow \mathbb{R} : \|f\|_{L^p(P_0)} < \infty\}$, where $\|f\|_{L^p(P_0)} := [\int |f|^p dP_0]^{1/p}$ for $p \in [1, \infty)$ and $\|f\|_{L^\infty(P_0)} := \text{ess sup}_v |f(v)|$. Our objective is to test hypotheses of the form:

$$H_0 : \Psi_{0,V} = \theta_0 \quad \text{versus} \quad H_1 : \Psi_{0,V} \neq \theta_0, \quad \text{almost surely} \quad (1)$$

where $\theta_0 = \int \Psi_{0,V}(v) dP_{0,V}(v)$. Notice that a special case of (1) is when θ_0 is equal to zero. Before presenting our general framework for testing (1), we first discuss two motivating examples that fall within this class: testing for conditional mean dependence and assessing the presence of treatment effect heterogeneity. In the Supplementary Material S2, we also include hypotheses tests about the conditional covariance function and conditional survival function as additional examples.

Example 1 (Testing for conditional mean dependence)

A fundamental task in predictive modeling is to determine whether a set of covariates can significantly improve the prediction of an outcome of interest in a mean-squared error sense. This is typically

assessed by studying the relationship between the outcome and covariates using the conditional mean function and determining whether the conditional mean function depends on the given set of covariates. When the conditional mean function is correctly specified as a linear function of the covariates, standard techniques such as the F -test provide valid and powerful tests for the significance of the covariates. However, when the linear model is misspecified, type-I error rate may not be controlled and power can be substantially reduced. To overcome challenges associated with model misspecification, [Zhang et al. \(2018\)](#) and [Lai et al. \(2021\)](#) proposed model-free tests for evaluating the null hypothesis of conditional mean independence. These methods improve power by formulating conditional mean independence as a global norm-based hypothesis on a functional object, enabling detection of broad classes of nonlinear alternatives. Despite these advances, overall conditional mean independence testing remains an active area of research, particularly in high-dimensional and nonparametric settings ([Hudson et al., 2026](#); [Li et al., 2023b](#); [Williamson et al., 2023](#); [Cai et al., 2024](#)). We show that hypothesis testing for conditional mean dependence fits within the general framework described in (1).

Let $O = (X, Y) \sim P_0$, where $Y \in \mathbb{R}$ is the outcome/response of interest, assumed to be bounded, and $X \in \mathbb{R}^p$ is a set of covariates. We define our smooth map as $\Psi(P) = \int y dP$ for $P \in \mathcal{M}$, then the function-valued and null parameters are $\Psi_{0,X} = \mathbb{E}[Y|X]$ and $\theta_0 = \mathbb{E}[Y]$, respectively. To evaluate the null hypothesis of conditional mean independence, we test that the conditional mean of Y given X does not depend on X . Specifically, the null hypothesis of interest is:

$$H_0 : \mathbb{E}[Y|X] = \mathbb{E}[Y] \quad \text{almost surely.}$$

The above falls within our general class of null hypotheses in (1).

Example 2 (Assessing presence of treatment effect heterogeneity)

Evaluating whether responses to an intervention are heterogeneous across a population has been of interest for many years (see [Byar, 1985](#), for early discussions). When heterogeneity arises among subgroups defined by discrete covariates (e.g., sex, or age groups), several parametric ([Byar, 1985](#); [Gail and Simon, 1985](#)) and nonparametric ([Imai and Li, 2024](#); [Dai et al., 2023](#); [Chernozhukov et al., 2017](#); [Crump et al., 2008](#)) tests for treatment effect heterogeneity have been proposed, mainly for randomized trials, with some extensions to observational studies. These methods assess effect heterogeneity by testing the constancy of the conditional average treatment effect (CATE), where the conditioning

covariate corresponds to the subgroup-defining variable. However, in many settings, investigators do not have clearly defined subgroups, but instead observe continuous or multivariate covariates that may capture the heterogeneity. Ad-hoc strategies, such as discretizing continuous covariates and applying the existing methods often yield low-powered tests, especially in small samples.

To test the null hypothesis of treatment effect homogeneity, we also propose evaluating the constancy of the CATE. Specifically, we test the null hypothesis that the CATE does not depend on the conditioning variable. The test that we propose does not require pre-defined subgroups and applies even when the conditioning covariate is continuous or multivariate.

Let $O = (Y^{(1)}, Y^{(0)}, T, X) \sim P_0$ be the data we wish to have observed, where $Y^{(t)}$ denote the potential outcome under treatment $T = t$ and X be baseline covariates. Let $T \in \{0, 1\}$ and for $s \subset \{1, \dots, p\}$, let X_s be the subvector of X containing elements with indices belonging to s . For this example, our smooth map is $\Psi(P) = \int Y(1) - Y(0)dP$ for $P \in \mathcal{M}$, then the function-valued and null parameters are $\Psi_{0, X_s} = \mathbb{E}[Y^{(1)} - Y^{(0)} | X_s]$ and $\theta_0 = \mathbb{E}[Y^{(1)} - Y^{(0)}]$, respectively. In this case, the null hypothesis of interest,

$$H_0 : \mathbb{E}[Y^{(1)} - Y^{(0)} | X_s] = \mathbb{E}[Y^{(1)} - Y^{(0)}] \quad \text{almost surely,}$$

also falls within our general framework described in (1). In the literature, the parameters θ_0 and Ψ_{0, X_s} are referred to as the ATE and CATE, respectively.

The two illustrative examples (Examples 1–2) represent many other statistical testing problems that can be motivated under the stated general form in (1). We will present detailed discussion of the application of the proposed test to these examples in Section 4 and evaluate finite sample performance through simulations in Section 6.

3 Proposed methodology

3.1 Testing strategy

In what follows, we develop a test for (1). Suppose $\Psi_{0, V} \in L^p(P_0)$ and θ_0 is finite. Then, the null hypothesis in (1) is equivalent to $\|\Psi_{0, V} - \theta_0\|_{L^p(P_0)} = 0$. When expressing the null hypothesis as a comparison between the norm and zero, one seemingly straightforward option to construct a test is to perform a Wald-type test based on an $n^{1/2}$ -consistent asymptotically normal estimator (Cai et al.,

2024; Dai et al., 2022; Williamson et al., 2021). However, these estimators do not admit to a tractable limiting null distribution due to the boundary null problem: under the null, the first-order influence function often vanishes, leading to a degenerate limiting null distribution of the test statistic (Verdinelli and Wasserman, 2024).

To address this challenge, we propose a test statistic motivated by an alternative representation of norms as supremum (via Hölder’s inequality) over a suitable function class (Folland, 1999, Proposition 6.13). The null hypothesis in (1) can be equivalently formulated as

$$H'_0 : \sup_{\substack{h \in L^q(P_0) \\ \|h\|_{L^q(P_0)}=1}} \left| \int \{ \Psi_{0,V}(v) - \theta_0 \} h(v) dP_{0,V}(v) \right| = 0, \quad (2)$$

where q is the conjugate exponent of p (i.e., $\frac{1}{p} + \frac{1}{q} = 1$) and $L^q(P_0)$ is the dual of $L^p(P_0)$. We illustrate this formulation for $p \in \{1, 2\}$. For $p = 1$, the maximizer of supremum in (2) is $h^* = \text{sign}\{\Psi_{0,V} - \theta_0\}$, yielding $\int \{ \Psi_{0,V} - \theta_0 \} \text{sign}\{\Psi_{0,V} - \theta_0\} dP_{0,V} = \int |\Psi_{0,V} - \theta_0| dP_0$. For $p = 2$, $h^* = (\Psi_{0,V} - \Psi_0) / \|\Psi_{0,V} - \Psi_0\|_{L^2(P_0)}$, and the supremum becomes $\int \{ \Psi_{0,V} - \theta_0 \} (\Psi_{0,V} - \theta_0) / \|\Psi_{0,V} - \theta_0\|_{L^2(P_0)} dP_{0,V} = [\int (\Psi_{0,V} - \theta_0)^2 dP_0]^{1/2}$.

Motivated by this dual representation, one could attempt to construct a test by estimating all linear transformations of $\Psi_{0,V} - \theta_0$ and explicitly evaluating the supremum over $L^q(P_0)$. However, this is technically challenging and practically infeasible, as $L^q(P_0)$ is too large. Instead, we restrict attention to a rich but non-complex function class $\mathcal{H}$ (e.g., one with bounded entropy) that captures meaningful deviations from the null. Specifically, defining $\bar{h}_0 := \int h(v) dP_{0,V}$, $h_{P_{0,V}}^c := h(V) - \bar{h}_0$, and $\Omega_{P_0}(h) := \int \Psi_{0,V} h_{P_{0,V}}^c(v) dP_{0,V}$ for $h \in \mathcal{H} \subset L^q(P_0)$, we consider the following null hypothesis

$$H''_0 : \sup_{h \in \mathcal{H}} |\Omega_{P_0}(h)| = 0. \quad (3)$$

From (3), we construct our test by choosing $T_n(\mathcal{H}) = \sup_{h \in \mathcal{H}} |n^{1/2} \hat{\Omega}_n(h)|$ as the test statistic, where $\hat{\Omega}_n(h)$ is an EIF-based estimator of $\Omega_{P_0}(h)$. This is attractive because, as shown later in Theorem 2, $T_n(\mathcal{H})$ converges weakly to a continuous transformation of a Gaussian process under some regularity conditions and hence has a tractable limiting null distribution. Moreover, if H_0 is true, then H''_0 is also true, although the converse does not necessarily hold: H''_0 being true does not necessarily imply that H_0 is true if $\mathcal{H}$ is not dense in $L^q(P_0)$. Nevertheless, the test based on (3) ensures valid type-I

error control for H_0 for any appropriate choice of $\mathcal{H}$, and it has power against alternatives aligned with the chosen class $\mathcal{H}$. The remainder of the paper describes the construction of the test in detail.

3.2 Theoretical foundation

For $P_0 \in \mathcal{M}$, let s be a function for which $\int s(o)dP_0(o) = 0$ and $\int s^2(o)dP_0(o) < \infty$. We define a path through P_0 in the direction s as $dP_t = (1 - ts)dP_0$. Similarly, for the conditional distribution of O given V fixed at v , under P_0 , we define a path through $P_{0,O|v}$ in the direction s_v as $dP_{t,O|v} = (1 - ts_v)dP_{0,O|v}$, where s_v is uniformly bounded and satisfies $\int s_v(o)dP_{0,O|v}(o) = 0$. We simplify the notation by writing $P_0 s := \int s(o)dP_0(o)$ and $\mathbb{P}_n s := \frac{1}{n} \sum_{i=1}^n s(O_i)$. To construct an EIF-based estimator of $\Omega_{P_0}(h) = \int \Psi_{0,V}(v) \{h(v) - \bar{h}_0\} dP_{0,V}(v)$ in the nonparametric model, we require $\Psi_{0,V}$ and Ψ_0 to satisfy the following regularity conditions:

C1: For some uniformly bounded function $D_{P_0} : \mathcal{O} \rightarrow \mathbb{R}$ such that $\int D_{P_0}(o)dP_0 = 0$,

$$\left. \frac{d\Psi_t}{dt} \right|_{t=0} = \int D_{P_0}(o)s(o)dP_0.$$

C2: For some uniformly bounded function $D_{P_0}^v : \mathcal{O} \rightarrow \mathbb{R}$ such that $\int D_{P_0}^v(o)dP_{0,O|v} = 0$, and

$$\left. \frac{d\Psi_{t,V}}{dt} \right|_{t=0} = \int D_{P_0}^v(o)s_v(o)dP_{0,O|v} \text{ holds } P_{0,V}\text{-almost surely.}$$

Condition [C1](#) requires that $\Psi(P)$ is a smooth finite-dimensional parameter, in the sense that $\Psi(P)$ is pathwise differentiable at P_0 relative to $\mathcal{M}$ with efficient influence function (EIF) D_{P_0} ([Bickel et al., 1993](#)). Condition [C2](#) is a conditional version of pathwise differentiability, in that, we take the Gâteaux derivative of $\Psi_{t,V}$ with respect to the conditional distribution of O given $V = v$ induced by P_0 . We will refer to $D_{P_0}^v$ as the conditional influence function with respect to $P_{0,O|v}$. Putting all together, we can characterize the EIF of $\Omega_{P_0}(h)$. We present this result in what follows.

Theorem 1. *Assume conditions [C1](#) and [C2](#) hold. Then, the functional $\Omega_{P_0}(h)$ is pathwise differentiable with efficient influence function*

$$D_{h,0}^*(o) = \left(D_{P_0}^v(o) - \int \Psi_{0,V}(v)dP_{0,V} + \Psi_{0,V} \right) h_{P_{0,V}}^c(v) - \Omega_{P_0}(h).$$

Theorem [1](#) shows that, under some regularity conditions, we can construct an estimator, $\hat{\Omega}_n(h)$,

of $\Omega_{P_0}(h)$ based on the EIF, $D_{h,0}^*$, that converges to a Gaussian distribution for a fixed h . Assuming further that $\mathcal{H}$ is P_0 -Donsker, we can extend this convergence to be uniform over $\mathcal{H}$. That is, $\left\{n^{1/2}(\widehat{\Omega}_n(h) - \Omega_{P_0}(h)) : h \in \mathcal{H}\right\}$ will converge to a mean-zero Gaussian process $\mathbb{G}$. Therefore, defining our test statistic as some continuous function of $n^{1/2}(\widehat{\Omega}_n(h))$, say $T_n(\mathcal{H}) = \sup_{h \in \mathcal{H}} \left|n^{1/2}(\widehat{\Omega}_n(h))\right|$, guarantees a well-defined limiting null distribution for our test, as well as theoretical size guarantees for testing H_0 .

3.3 Calculating the conditional influence function

Obtaining $D_{h,0}^*$ from Theorem 1 requires deriving the conditional influence function, $D_{P_0}^v$, associated with $\Psi_{0,V}$. Therefore, the derivation of $D_{h,0}^*$ is problem-dependent, often requiring knowledge of functional derivatives. In what follows, we show that $D_{P_0}^v$ can be readily obtained for certain classes of function-valued parameters. In particular, in many instances, one can readily obtain $D_{P_0}^v$ from the form of the EIF of Ψ_0 , which is often available from existing results.

The first class of function-valued parameters we discuss consist of those for which the smooth map $\Psi(P)$ can be expressed as a continuous linear functional of the conditional mean function. Let $\mu_0(v) = \mathbb{E}[U \mid V = v]$ denote the conditional mean function, and let $m(O, \mu_0)$ be a continuous linear functional of μ_0 . Then $\Psi(P) = \int m(O, \mu_0) dP$, with $\Psi_0 = \mathbb{E}[m(O, \mu_0)]$ and $\Psi_{0,V} = \mathbb{E}[m(O, \mu_0) \mid V = v]$. As established in Chernozhukov et al. (2018b), if $\mathbb{E}[m(O, \mu_0)] = \mathbb{E}[a_0(O)\mu_0]$ for some function a_0 satisfying $\mathbb{E}[a_0(O)] < \infty$, then the EIF of Ψ_0 takes the form

$$D_{P_0}(o) = m(o, \mu_0) - \Psi_0 + a_0(o)\{u - \mu_0(v)\}.$$

This result covers a broad class of statistical functionals that have been well studied in the literature. We summarize in Proposition 1 the form of the EIF of $\Omega_{P_0}(h)$, $D_{h,0}^*$, under the assumption that the map $\Psi(P)$ is a continuous linear functionals of the conditional mean function.

Proposition 1. *Let $a_0 : \mathcal{O} \rightarrow \mathbb{R}$ such that $\mathbb{E}[a_0(O)^2] < \infty$ and suppose*

$$\Psi_{0,V} = \mathbb{E}[m(O, \mu_0) \mid V] = \mathbb{E}[a_0(O)\mu_0(V) \mid V], \text{ for all } \mu_0 \text{ with } \mathbb{E}[\mu_0(V)^2] < \infty.$$

Then, the EIF of $\Omega_{P_0}(h)$, $D_{h,0}^*(o)$, simplifies to

$$\left(m(o, \mu_0) + a_0(o)[u - \mu_0(v)] - \int \Psi_{0,V}(v) dP_{0,V} \right) h_{P_{0,v}}^c(v) - \Omega_{P_0}(h).$$

We show in Section 4 that Examples 1 and 2 from Section 2 fall within the class of continuous linear functionals of the conditional mean function; hence, the EIF can be obtained directly using Proposition 1. The conditional covariance functional, discussed in the Supplementary Material S2, does not belong to this class. In what follows, we introduce another broad class of functionals that includes the conditional covariance functional. Consider the smooth map, $\Psi(P) = \int g(O; \cdot) dP$, where g depends on P_0 only through derived marginal or conditional distributions under P_0 . Proposition 2 shows that for function-valued parameters of the form $\mathbb{E}[g(O; P_{0,O|V})|V]$, the EIF of $\Omega_{P_0}(h)$ admits a conditional object replacement analogous to that in Proposition 1. The following result formalizes this statement.

Proposition 2. *Define the smooth map, $\Psi(P) = \int g(O; \cdot) dP$, where g depends on P_0 only through derived marginal or conditional distributions under P_0 . Suppose the finite-dimensional parameter $\Psi_0 = \mathbb{E}[g(O; P_{0,u})]$ is pathwise differentiable at $P_{0,u}$ relative to the nonparametric model $\mathcal{M}_u$ (all distributions on $\mathcal{U}$), with EIF $D_{P_{0,u}}$. For the function-valued parameter $\Psi_{0,V} = \mathbb{E}[g(O; P_{0,O|V})|V = v]$, if there exists $D_{P_0}^v$ such that Condition C2 holds, then the EIF of $\Omega_{P_0}(h)$, denoted $D_{h,0}^*(o)$, simplifies to*

$$D_{h,0}^*(o) = \left(D_{P_{0,O|v}}(o) - \mathbb{E}[g(O; P_{0,O|V})] + \mathbb{E}[g(O; P_{0,O|V})|V = v] \right) h_{P_{0,V}}^c(v) - \Omega_{P_0}(h). \quad (4)$$

Here, $D_{P_{0,O|v}}$ is obtained by replacing $P_{0,u}$ in the expression for $D_{P_{0,u}}$ with $P_{0,O|v}$.

Similar to Proposition 1, this result shows that $D_{h,0}^*$ can be obtained without explicitly deriving $D_{P_0}^v$ from first principles. Function-valued parameters within this class include the conditional mean function, conditional variance, conditional expected density, and conditional covariance. While the conditional mean function belongs to both classes, the CATE belongs only to the class described in Proposition 1. Together, Propositions 1 and 2 provide a general strategy for obtaining the EIF, $D_{h,0}^*$, across a wide range of problems by replacing unconditional objects with their conditional counterparts in known efficient influence functions.

Remark 1. *The conclusion from Proposition 1 extends to a broader class of function-valued parameters beyond those that can be expressed as bounded linear functionals of the conditional mean.*

Specifically, this result can be extended to the class of functionals of the form $\Psi_0 = \mathbb{E}[m(O, \eta_0)]$ and $\Psi_{0,V} = \mathbb{E}[m(O, \eta_0)|V]$, where $m(o, \eta_0)$ is a smooth function of some summary η_0 of the conditional distribution. If the conditional influence function for η_0 exist, then one can obtain the conditional influence function of $\Psi_{0,V}$ from the form of the efficient influence function of Ψ_0 . If η_0 is the conditional mean, this result reduces to Proposition 1. However, η_0 can be any conditional quantity for which we can obtain its conditional influence function. An example is when η_0 is the conditional survival function. We present this general result in the Supplementary Material S2 and derive the form of $D_{P_0}^v$ for when η_0 is the conditional survival function.

3.4 Test statistic and its limiting null distribution

We propose our test statistic as a function of an estimator of $\Omega_{P_0}(h)$. Importantly, we want the estimator to be flexible but also attains parametric-rate of convergence. To this end, we introduce the naive plug-in estimator, from which we would build an estimator that achieves $n^{1/2}$ -rate of convergence. Let $\hat{P}_n \in \mathcal{M}$ be an estimate of P_0 based on the sample $O_1, O_2, \dots, O_n$. We construct a plug-in estimator for the parameter $\Omega_{P_0}(h)$ as

$$\hat{\Omega}_{\hat{P}_n}(h) = n^{-1} \sum_{i=1}^n \Psi_{n,V}(V_i) \left[h(V_i) - n^{-1} \sum_{j=1}^n h(V_j) \right],$$

where $\Psi_{n,V}$ is a nonparametric estimator of $\Psi_{0,V}$. Although the plug-in estimator is simple to obtain, it generally inherits bias from estimating nuisance functions with flexible nonparametric estimators. To correct the bias of the plug-in estimator, we propose to use the one-step bias correction method (Pfanzagl and Wefelmeyer, 1985) by adding a correction term based on the EIF of $\Omega_{P_0}(h)$. Under some regularity conditions, the de-bias estimator retains a normal asymptotic limit. We define the one-step estimator $\Omega_n^{os}(h)$ at a fixed h as

$$\begin{aligned} \hat{\Omega}_n^{os}(h) &= \hat{\Omega}_{\hat{P}_n}(h) + n^{-1} \sum_{i=1}^n D_{h,n}^*(O_i) \\ &= n^{-1} \sum_{i=1}^n \left[D_n^v(O_i) - n^{-1} \sum_{i=1}^n \Psi_{n,V}(V_i) + \Psi_{n,V}(V_i) \right] \left(h(V_i) - n^{-1} \sum_{j=1}^n h(V_j) \right). \end{aligned}$$

Here, $\Psi_{n,V}$ and D_n^v could depend on other nuisance functions (e.g., the conditional mean function) that are specific to the function-valued paramter in the testing problem.

Algorithm 1 Bootstrap test procedure

Require: Observed data $\{O_i\}_{i=1}^n$, estimator $D_{h,n}^*(o)$ for efficient influence function, number of bootstrap samples B , significance level α

1: **For** $b = 1, \dots, B$, do:

2: Draw independent standard normal variables $\xi_1^{(b)}, \dots, \xi_n^{(b)}$

3: Compute the multiplier bootstrap statistic: $T^{(b)}(\mathcal{H}) = \sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^n \xi_i^{(b)} D_{h,n}^*(O_i) \right|$

4: Compute the observed test statistic: $T_n(\mathcal{H}) = \sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^n D_{h,n}^*(O_i) \right|$

5: Compute the bootstrap p -value:

$$\rho_B = \frac{1}{B} \sum_{b=1}^B \mathbb{1} \left\{ T^{(b)}(\mathcal{H}) > T_n(\mathcal{H}) \right\}$$

6: **Reject** H_0 at level α if $\rho_B \leq \alpha$.

In Theorem (2), we state sufficient conditions under which $\{\widehat{\Omega}_n^{os} : h \in \mathcal{H}\}$ has a tractable limiting distribution, thereby ensuring asymptotic validity of our proposed testing procedure. In particular, we provide conditions under which $\widehat{\Omega}_n^{os}$ is uniformly asymptotically linear with influence function $D_{h,0}^*$ and has a Gaussian limit; that is,

$$\widehat{\Omega}_n^{os}(h) - \Omega_0(h) = \sum_{i=1}^n D_{h,n}^*(O_i) + r_n(h), \quad (5)$$

where $\sup_{h \in \mathcal{H}} |r_n(h)| = o_P(n^{-1/2})$, and the standardized process $\{n^{1/2}[\widehat{\Omega}_n^{os}(h) - \Omega_0(h)] : h \in \mathcal{H}\}$ is asymptotically Gaussian.

Theorem 2. *Suppose the following conditions hold: i) $\mathcal{H}$ is a P_0 -Donsker class, ii) $\{D_{h,n}^*, D_{h,0}^* : h \in \mathcal{H}\}$ belong to a P_0 -Donsker class, (iii) $\sup_{h \in \mathcal{H}} |r_n(h)| = o_P(n^{-1/2})$, where $r_n(h)$ is defined in (5). Then, $\{n^{1/2}[\widehat{\Omega}_n^{os}(h) - \Omega_0(h)] : h \in \mathcal{H}\}$ converges weakly in the space $\ell^\infty(\mathcal{H})$ of bounded real-valued functions on $\mathcal{H}$ to a mean-zero Gaussian process $\mathbb{G}$ with covariance function $\Sigma_0(h_1, h_2) = P_0 \left[D_{h_1,0}^* D_{h_2,0}^* \right]$.*

Conditions (i) and (ii) restrict the complexity of the nuisance functions, their estimators, and the function class $\mathcal{H}$. Donsker classes are often quite large, so the assumption is not overly restrictive. Moreover, the Donsker condition is satisfied by a broad range of function classes, including those with bounded variation norm. Nonparametric estimators can also be used, and many are known to satisfy these properties under weak conditions. Examples include penalized estimators and sieve-based procedures (Geer, 2000). For other function classes, the Donsker assumption can be verified by analyzing the metric entropy of the function class (Van der Vaart, 2000; Kosorok, 2008). Finally,

condition *iii*) pertains to the remainder term, ensuring that the estimator is uniformly asymptotically linear. This condition is often met by the one-step estimator, which we illustrate through specific examples in Section 4.

Leveraging Theorem 2, we choose as our test statistic $T_n(\mathcal{H}) := \sup_{h \in \mathcal{H}} n^{1/2} |\hat{\Omega}_n^{os}(h)|$. By the continuous mapping theorem, $\sup_{h \in \mathcal{H}} n^{1/2} |\hat{\Omega}_n^{os}(h)|$ weakly converges to $\sup_{h \in \mathcal{H}} |\mathbb{G}(h)|$ under the null. Therefore, a test that rejects the null when $T_n(\mathcal{H})$ exceeds the $1 - \alpha$ quantile of $\sup_{h \in \mathcal{H}} |\mathbb{G}(h)|$ achieves type-1 control at the level α . While integrating under the exact distribution of $\sup_{h \in \mathcal{H}} |\mathbb{G}(h)|$ is difficult, we can use an approximation instead. To approximate the limiting process $\sup_{h \in \mathcal{H}} |\mathbb{G}(h)|$, we draw a B set of n *i.i.d.* samples from a distribution with mean zero and unit variance. For each $b = 1, \dots, B$, denote by $\xi_1^{(b)}, \xi_2^{(b)}, \dots, \xi_n^{(b)}$ the set of n random samples from a mean zero and unit variance distribution. We calculate the multiplier bootstrap statistic as $T^{(b)}(\mathcal{H}) = \sup_{h \in \mathcal{H}} |n^{-1/2} \sum_{i=1}^n \xi_i D_{h,n}^*(O_i)|$. The so-called multiplier bootstrap samples $T^{(1)}(\mathcal{H}), T^{(2)}(\mathcal{H}), \dots, T^{(m)}(\mathcal{H})$ approximate the distribution of $\sup_{h \in \mathcal{H}} |\mathbb{G}(h)|$ for n and B large. See Hsu (2016) and Hudson et al. (2026) for discussions of such simulated processes. The detailed procedure for constructing the test for null hypotheses (3) is presented in Algorithm 1.

4 Illustrations

In this section, we show how our proposed framework applies to Examples 1–2 introduced in Section 2. Specifically, for each example, we present the EIF, $D_{h,0}^*$, and one-step estimator, $\hat{\Omega}_n^{os}(h)$ of $\Omega_{P_0}(h)$. We defer discussion of conditions under which $\sup_{h \in \mathcal{H}} |r_n(h)| = o_p(n^{-1/2})$ and detailed derivation of the presented results to the Supplementary Material S3.

Example 1 (Continued)

Recall that the null hypothesis for testing conditional mean independence is

$$H_0 : \mathbb{E}[Y|X] = \mathbb{E}[Y]$$

Here, $\theta_0 = \mathbb{E}[Y]$, $\Psi_{0,X} = \mathbb{E}[Y|X]$, and $\Omega_{P_0}(h) = \int \mathbb{E}[Y|X = x] h_{P_{0,X}}^c(x) dP_{0,X}$. The EIF for $\Omega_{P_0}(h)$ can be characterized using Proposition 1. For $\mu_0(x) = \mathbb{E}[Y|X = x]$, $a_0(o) = 1$ and $m(o, \mu_0) = \mu_0$, we can express $\Psi_{0,X}$ as a continuous linear functional of the conditional mean function. That is, $\Psi_{0,X} = \mathbb{E}[m(o, \mu_0)|X]$. Hence, following Proposition 1, $D_{P_0}^v = y - \mathbb{E}[Y|X = x]$, and the EIF, $D_{h,0}^*$,

and one-step estimator, $\widehat{\Omega}_n^{os}(h)$, of $\Omega_{P_0}(h)$ are given by

$$D_{h,0}^*(o) = (y - \mathbb{E}[Y])h_{P_0,X}^c(x) - \Omega_{P_0}(h),$$

$$\widehat{\Omega}_n^{os}(h) = n^{-1} \sum_{i=1}^n (Y_i - \bar{Y}_n)h_n^c(X_i),$$

where $h_n^c(X) = h(X) - \bar{h}_n$, and $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i$, and $\bar{h}_n = \frac{1}{n} \sum_{i=1}^n h(X_i)$ are estimators of $\mathbb{E}[Y]$, and $\mathbb{E}[h(X)]$, respectively.

Example 2 (Continued)

The null hypothesis for testing treatment effect heterogeneity is

$$H_0 : \mathbb{E}[Y^{(1)} - Y^{(0)}|X_s] = \mathbb{E}[Y^{(1)} - Y^{(0)}].$$

Under standard causal assumptions of consistency, positivity, and conditional exchangeability ([Hernán and Robins, 2020](#)), the CATE and ATE can be identified in both randomized and observational studies as $\mathbb{E}[\mathbb{E}[Y|T = 1, X] - \mathbb{E}[Y|T = 0, X]|X_s]$ and $\mathbb{E}[\mathbb{E}[Y|T = 1, X] - \mathbb{E}[Y|T = 0, X]]$, respectively. Letting $\mu_0(t, x) = \mathbb{E}[Y|T = t, X = x]$ and $\pi_0(t|x) = \Pr(T = t|X = x)$, we denote $\tau_{0,X_s} = \mathbb{E}[\mu_0(1, x) - \mu_0(0, x)|X_s]$ and $\tau_0 = \mathbb{E}[\mu_0(1, x) - \mu_0(0, x)]$ as the identified CATE and ATE, respectively. Here, $\Psi_{0,X_s} = \mathbb{E}[\mu_0(1, X) - \mu_0(0, X)|X_s]$, $\theta_0 = \mathbb{E}[\mu_0(1, X) - \mu_0(0, X)]$, and $\Omega_{P_0}(h) = \int \mathbb{E}[\mu_0(1, x) - \mu_0(0, x)|X_s = x]h_{P_0,X_s}^c(x_s) dP_{0,X_s}$. Similar to Example 1, we can express Ψ_{0,X_s} as a continuous linear functional of the conditional mean function. To see this, let $a_0(o) = \frac{t}{\pi_0(t|x)} - \frac{1-t}{1-\pi_0(t|x)}$ and $m(x, \mu_0) = \mu_0(1, x) - \mu_0(0, x)$. Then, we have $\Psi_{0,X_s} = \mathbb{E}[a_0(O)\mu_0(T, X)|X_s]$. Hence, following Proposition 1, $D_{P_0}^{x_s}(o) = m(x, \mu_0) + a_0(x)[y - \mu_0(t, x)]$. Thus, the EIF, $D_{h,0}^*$ and one-step estimator $\widehat{\Omega}_n^{os}(h)$ of $\Omega_{P_0}(h)$ are given by

$$D_{h,0}^*(o) = (\mu_0(1, x) - \mu_0(0, x) + a_0(x)\{y - \mu_0(a, x)\} - \tau_0 + \tau_{0,X_s})h_{P_0,X_s}^c(x_s) - \Omega_{P_0}(h),$$

$$\widehat{\Omega}_n^{os}(h) = n^{-1} \sum_{i=1}^n (\mu_n(1, X_i) - \mu_n(0, X_i) + a_n(X_i)\{Y_i - \mu_n(T_i, X_i)\} - \tau_n + \tau_{n,X_s}(O_i)) (h(X_{s,i}) - \bar{h}_n),$$

where we have replaced the subscript “0” with “n” to denote an estimator of the respective parameter (i.e., $\mu_n(t, x)$ is the estimator of $\mu_0(t, x)$).

5 Aggregate test to improve power

So far, we have discussed our inferential strategy assuming that one has knowledge of the function class $\mathcal{H}$. In practice, no single choice of function class can guarantee good power across all alternatives. In fact, the optimal choice of function class for ensuring good power against specific alternatives is typically unknown. To ensure reasonable power against a broader class of alternatives, we propose constructing an aggregate test that combines test statistics across different pre-specified function classes.

Let $L \in \mathbb{N}$ be a fixed natural number, and $\{\mathcal{H}_\ell\}_{\ell=1}^L$ denote a collection of function classes. For each ℓ , let $T_n(\mathcal{H}_\ell)$ be our test statistic constructed using function class $\mathcal{H}_\ell$, and define the L -dimensional vector of observed test statistics $\mathbf{T}_n := (T_n(\mathcal{H}_1), \dots, T_n(\mathcal{H}_L)) \in \mathbb{R}^L$. To ease notations, we simply denote $\mathbf{T}_n^{(0)}$ by $\mathbf{T}^{(0)}$. Suppose we generate B bootstrap samples from the limiting null distribution of $\mathbf{T}^{(0)}$. Let $\mathbf{T}^{(b)} = \{T^{(b)}(\mathcal{H}_\ell)\}_{\ell=1}^L$ be the vector of size L corresponding to the b^{th} bootstrap sample, and $\mathbf{T}^{(-b)}$ denote the collection of vectors obtained by excluding $\mathbf{T}^{(b)}$ but including the observed statistic $\mathbf{T}^{(0)}$. We include $\mathbf{T}^{(0)}$ in $\mathbf{T}^{(-b)}$ to preserve exchangeability between the observed and bootstrap samples under the null. Our proposed aggregate test is based on assessing how extreme a scalar aggregate summary of $\mathbf{T}_n^{(0)}$ is relative to samples from a bootstrap approximation of the summary's sampling distribution. Algorithm 2 present detailed steps for performing an aggregate test when $\mathbf{T}_n^{(0)}$ is summarized using a scaled and centered ℓ_2 norm. A similar strategy is also presented in [Shah and Bühlmann \(2018\)](#), which combines multiple test statistics obtained from multiple tuning parameters in a penalized high-dimensional regression setting. Alternative aggregating functions may also be used, including the absolute maximum and the ℓ_1 norm.

The resulting p -value, from the proposed aggregate test is valid in the sense that $\Pr(P_{\text{agg}} \leq \alpha \mid H_0) \leq \alpha$, since under the null hypothesis the collection $\{\mathbf{T}_n^{(0)}, \mathbf{T}^{(1)}, \dots, \mathbf{T}^{(B)}\}$ is exchangeable, the ℓ_2 norm is an order-preserving transformation that preserves exchangeability, and standard results for Monte Carlo tests therefore imply validity ([Davison and Hinkley, 1997](#); [Shah and Bühlmann, 2018](#)). Other order-preserving continuous aggregation functions such as the arithmetic mean or the maximum could also be used in place of the squared average employed here.

In the next section, we present concrete examples of $\mathcal{H}$ and discuss their merits and limitations. It is important that one chooses $\mathcal{H}$ large enough that whenever H_0 fails, there exist some $h^* \in \mathcal{H}$ with $|\Omega_{P_0}(h^*)| > 0$. We discuss two choices of $\mathcal{H}$ that are popular for their computational and statistical

Algorithm 2 Aggregate test procedure

Require: Observed vector of test statistics $\mathbf{T}_n^{(0)} = \{T_n(\mathcal{H}_\ell)\}_{\ell=1}^L$; B bootstrap samples $\{\mathbf{T}^{(b)}\}_{b=1,\dots,B}$; significance level α .

- 1: Given the B bootstrap samples $\{\mathbf{T}^{(b)}\}_{b=1,\dots,B}$ and observed test statistic $\mathbf{T}^{(0)}$, calculate $\hat{\boldsymbol{\mu}}^{(-b)} = (\hat{\mu}_1^{(-b)}, \dots, \hat{\mu}_L^{(-b)})$ and $\hat{\boldsymbol{\sigma}}^{(-b)} = (\hat{\sigma}_1^{(-b)}, \dots, \hat{\sigma}_L^{(-b)})$ as

$$\hat{\mu}_\ell^{(-b)} = \frac{1}{B} \sum_{\substack{b'=0 \\ b' \neq b}}^B T^{(b')}(\mathcal{H}_\ell), \quad \hat{\sigma}_\ell^{(-b)} = \left[\frac{1}{B-1} \sum_{\substack{b'=0 \\ b' \neq b}}^B (T^{(b')}(\mathcal{H}_\ell) - \hat{\mu}_\ell^{(-b)})^2 \right]^{1/2}.$$

- 2: **For each bootstrap sample** $b = 1, \dots, B$, do:

- 3: Compute the b^{th} bootstrap transformed value using the aggregate function

$$\tilde{Q}_b = \frac{1}{L} \sum_{\ell=1}^L \left(\frac{T^{(b)}(\mathcal{H}_\ell) - \hat{\mu}_\ell^{(-b)}}{\hat{\sigma}_\ell^{(-b)}} \right)^2.$$

- 4: Compute the observed aggregate test statistic

$$\tilde{Q}_0 = \frac{1}{L} \sum_{\ell=1}^L \left(\frac{T_n(\mathcal{H}_\ell) - \hat{\mu}_\ell}{\hat{\sigma}_\ell} \right)^2.$$

where $\hat{\mu}_\ell$ and $\hat{\sigma}_\ell$ correspond to $\hat{\mu}_\ell^{(-b)}$ and $\hat{\sigma}_\ell^{(-b)}$ for when $b = 0$.

- 5: Calculate aggregate p -value; $P_{\text{agg}} = \frac{1}{B+1} \left(1 + \sum_{b=1}^B \mathbb{I} \{ \tilde{Q}_b \geq \tilde{Q}_0 \} \right)$.

- 6: **Reject** H_0 at level α if $P_{\text{agg}} \leq \alpha$.
-

properties, namely, the thresholded indicator function class and the reproducing kernel Hilbert space. Other choices of function class can be seamlessly incorporated into the proposed aggregate testing framework.

5.1 Thresholded indicator

First, recall that for $p = 1$, the maximizer of (2) among the class of bounded functions is $h^* = \text{sign}\{\Psi_{0,V}\}$ up to constant proportionality. A simple function class that can approximate bounded functions on $\mathcal{V}$ is the class of thresholded indicators defined on each value in $\mathcal{V}$. In fact, if $\Psi_{0,V}$ is monotone on $\mathcal{V}$, the maximizer, h^* , belongs to this class. We define the class of thresholded indicators on $\mathcal{V}$ as $\mathcal{H} = \{h_{v_0}(v) = \mathbb{1}\{v \leq v_0\} : v_0 \in \mathcal{V}\}$. This choice of $\mathcal{H}$ is P_0 -Donsker (Van der Vaart, 2000),

and the corresponding test statistic is

$$T_n(\mathcal{H}) = \max_{v_0 \in \mathcal{V}} \left| \frac{1}{n} \sum_{i=1}^n \left[D_n^v(O_i) - \sum_{j=1}^n \Psi_{n,V}(V_j) + \Psi_{n,V}(V_i) \right] \left\{ \mathbb{1}\{V_i \leq v_0\} - \sum_{k=1}^n \mathbb{1}\{V_k \leq v_0\} \right\} \right|.$$

This class is appealing due to its computational simplicity and ease of implementation. However, when $\Psi_{0,V}$ is highly complex (e.g., sinusoidal or quadratic), choosing $\mathcal{H}$ as simple as the class of thresholded indicators may result in a loss of power of the test.

5.2 Reproducing kernel Hilbert space

We introduce an alternative function class, the reproducing kernel Hilbert space (RKHS). This function class is quite rich and provides a convenient way to approximate the norm in (2) for when $p = 2$. This means selecting $\mathcal{H}$ as the RKHS to construct our test may lead to a powerful test even when $\Psi_{0,V}$ is non-monotone. We consider the RKHS induced by the eigenfunctions $\psi_{2j-1}(x) = \sqrt{2} \cos(2j\pi x)$ and $\psi_{2j}(x) = \sqrt{2} \sin(2j\pi x)$ with their corresponding eigenvalues $\lambda_{2j-1} = \lambda_{2j} = (2j\pi)^{-4}$ for $j = 1, 2, \dots$. Each function, h , in the space can be expressed as $h(x) = \sum_{j=1}^{\infty} a_j \psi_j(x)$ such that its norm satisfies $\sum_{j=1}^{\infty} \frac{a_j^2}{\lambda_j} < \infty$ for $a_j \in \mathbb{R}$. Because the eigenvalues decay as $\lambda_j \asymp j^{-4}$, the coefficients a_j must decay sufficiently fast for the norm to remain finite. Consequently, the behavior of functions in $\mathcal{H}$ is largely determined by the low-frequency components of the expansion, and the contribution of higher-order terms becomes increasingly negligible. This motivates approximating h using only the first D basis functions for a sufficiently large integer D , which provides a good approximation to functions in $\mathcal{H}$. Let Γ denote a $D \times D$ diagonal matrix with entries $\Gamma_{j,j} = 1/\lambda_j$, and let $\mathbf{V}$ denote a $D \times D$ matrix with entries given by estimates of the covariance between each pair of eigenfunctions. We define the RKHS norm and variance for a function, $h(x) = \sum_{j=1}^D a_j \psi_j(x)$, as $\mathbf{a}^T \Gamma \mathbf{a}$ and $\mathbf{a}^T \mathbf{V} \mathbf{a}$, respectively. To restrict attention to functions that balance smoothness and variability, we consider the following constrained function class

$$\mathcal{H} = \left\{ \sum_{j=1}^D a_j \psi_j(x) : a_j \in \mathbb{R}, \gamma \mathbf{a}^T \Gamma \mathbf{a} + (1 - \gamma) \mathbf{a}^T \mathbf{V} \mathbf{a} \leq C, \text{ for some fixed } C > 0, \text{ and } \gamma \in [0, 1] \right\},$$

where $\mathbf{a} = (a_1, a_2, \dots, a_D)$. The parameter γ controls the trade-off between smoothness and variability: values of γ close to one place greater weight on the RKHS norm $\mathbf{a}^T \Gamma \mathbf{a}$ and therefore enforce smoother

functions, while values close to zero place greater weight on the variance term $\mathbf{a}^T \mathbf{V} \mathbf{a}$ and therefore restrict the variability of the function; the constant C determines the scale of the constraint and does not affect the shape of the function class. See Supplementary Material S1 for a more detailed discussion on the RKHS and motivation for the constraint on the function class.

A key advantage of using the RKHS is that we can explicitly express the one-step estimator, $\widehat{\Omega}_n^{\text{os}}(h)$, of $\Omega_{P_0}(h)$ as a linear function in $\mathbf{a}$. That is,

$$\begin{aligned}\widehat{\Omega}_n^{\text{os}}(h) &= \frac{1}{n} \sum_{i=1}^n \left[D_n^v(O_i) - \frac{1}{n} \sum_{k=1}^n \Psi_{n,V}(V_k) + \Psi_{n,V}(V_i) \right] \left(\sum_{j=1}^D a_j \psi_j(V_i) - \frac{1}{n} \sum_{k=1}^n \sum_{j=1}^D a_j \psi_j(V_k) \right) \\ &= \mathbf{U}_n^T \mathbf{a},\end{aligned}$$

where $\mathbf{U}_n = \mathbf{U}(\phi_n^*, \psi) = n^{-1} \Phi^T \phi_n^*$ is a D -dimensional vector. Here ϕ_n^* is an n -dimensional vector whose i th entry is $D_n^v(O_i) - \frac{1}{n} \sum_{k=1}^n \Psi_{n,V}(V_k) + \Psi_{n,V}(V_i)$, and Φ is an $n \times D$ matrix whose j th column has entries $\psi_j(V_i) - \frac{1}{n} \sum_{k=1}^n \psi_j(V_k)$ for $i = 1, 2, \dots, n$. Given the simplification above, we calculate our test statistic by performing the following maximization

$$T_n(\mathcal{H}) = \max_{\mathbf{a} \in \mathbb{R}^D} \left\{ n^{1/2} \mathbf{U}_n^T \mathbf{a} : \gamma \mathbf{a}^T \Gamma \mathbf{a} + (1 - \gamma) \mathbf{a}^T \mathbf{V} \mathbf{a} \leq C \right\}. \quad (6)$$

The solution to the maximization problem is obtained by maximizing the Lagrangian associated with (6). That is,

$$\hat{\mathbf{a}}_{\eta, \gamma} = \operatorname{argmax}_{\mathbf{a} \in \mathbb{R}^D} \left\{ n^{1/2} \mathbf{U}_n^T \mathbf{a} - \frac{\eta}{2} (\gamma \mathbf{a}^T \Gamma \mathbf{a} + (1 - \gamma) \mathbf{a}^T \mathbf{V} \mathbf{a}) \right\}, \quad (7)$$

where $\eta > 0$ is chosen to satisfy the constraint in (6). For fixed η , the solution to (7) is

$$\hat{\mathbf{a}}_{\eta, \gamma} = n^{1/2} \eta^{-1} (\gamma \Gamma + (1 - \gamma) \mathbf{V})^{-1} \mathbf{U}_n.$$

We can therefore express our test statistic as

$$T_n(\mathcal{H}) = \eta^{-1} n^{1/2} \mathbf{U}_n^T (\gamma \Gamma + (1 - \gamma) \mathbf{V})^{-1} n^{1/2} \mathbf{U}_n.$$

The Lagrange multiplier η primarily affects the scale of the test statistic but does not influence

the shape of its sampling distribution. For this reason, we typically fix η at a positive constant. In contrast, the tuning parameter γ governs the trade-off between smoothness and variance of the estimated functions and directly influences the sampling distribution of the test statistic. Consequently, each value of γ indexes a distinct function class and yields a corresponding test statistic with its own sampling distribution.

6 Numerical studies

In this section, we investigate the finite sample performance of the proposed test in the context of the two working examples introduced in Section 2. For each example, we study various settings under the null and alternative hypotheses. We implement our test using the thresholded indicator and RKHS as function classes. We truncate the basis expansion at $D = 100$ for the RKHS function class. From the RKHS, we construct $K = 50$ function classes based on a grid of 50 different γ values, $\gamma_{max} \left[\frac{\gamma_{min}}{\gamma_{max}} \right]^{\frac{k-1}{K-1}}$, where $k = 1, 2, \dots, K$, $\gamma_{max} = 1 \times 10^{-3}$, and $\gamma_{min} = 1 \times 10^{-5}$. We implement the aggregate test by combining the thresholded indicator and the K RKHS function classes. We approximate the null limiting distribution using $B = 800$ multiplier bootstrap samples.

Additionally, we consider the Cauchy combination method (Liu and Xie, 2020), an alternative approach for combining multiple tests. We refer to this test as the Cauchy test, or simply Cauchy. To investigate the advantages of combining different function classes, we also implement our procedure using individual function classes only, such as the thresholded indicators and the RKHS with only a smoothing penalty (corresponding to $\gamma = 1$), as well as an aggregate test combining only the 50 RKHS function classes. We refer to the test based on the thresholded indicators as Indicator test, the RKHS with only a smoothing penalty as FixedRKHS test, and the test that combines the 50 RKHS function classes as CombinedRKHS test. For Examples 1 and 2, we compare the performance of the proposed tests with tests specifically designed for these problems. We conduct 500 Monte Carlo simulations with sample sizes $n \in \{125, 250, 500, 1000, 2000\}$ across all settings.

Example 1: Testing for conditional mean dependence

For $X \sim \text{Unif}(-1, 1)$ and $\varepsilon \sim N(0, 1)$ that is independent of X , we consider three conditional outcome distributions:

- Setting 1: $Y = \varepsilon$

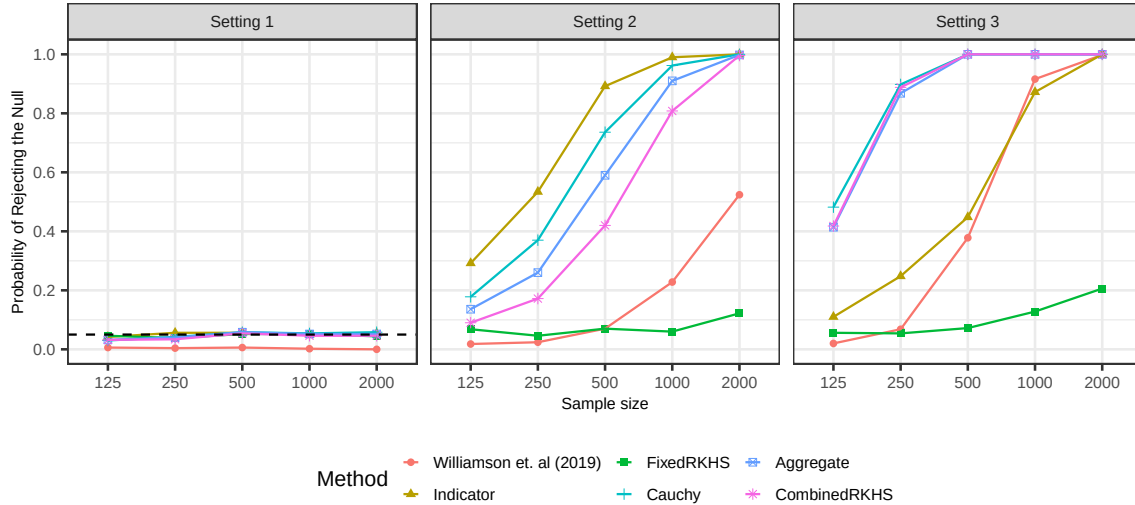


Figure 1: Empirical probability of rejection for hypotheses in Example 1 under different data generating mechanisms, at different sample sizes. Horizontal wide-dash line indicates the nominal 0.05 test size.

- Setting 2: $Y = 0.25X + \varepsilon$
- Setting 3: $Y = \sin(\pi X \text{sign}(X)) + \varepsilon$

For each setting, we compare the performance of our proposed tests with the approach by [Williamson et al. \(2023\)](#) implemented in the R package `vimp` ([Williamson, 2023](#)). This approach assesses whether the expected least squared error loss resulting from predicting Y using the conditional mean given X , falls below the marginal variance of Y .

The results are presented in Figure 1. Under Setting 1, where the null hypothesis $\mathbb{E}[Y|X] = \mathbb{E}[Y]$ holds, we observe that all the tests under our proposed framework are able to control the type-I error. However, the method by [Williamson et al. \(2023\)](#) appears to be conservative. Under Setting 2, where the alternative is linear, and hence monotone, the Indicator test has the highest power. We observe slower increase in power as n increases for the Cauchy and Aggregate tests since they are based on RKHS, which accommodates more complex functions. Nonetheless, the Aggregate and Cauchy tests are still able to achieve high power similar to the Indicator test for larger sample sizes. In Setting 3, where the conditional mean function is more complex compared to Setting 2, the Aggregate and Cauchy tests achieve high power at small sample size. The power of the Indicator test, on the other hand, increases more slowly with sample size. The power of the test of [Williamson et al. \(2023\)](#) increases with sample size in both Settings 2 and 3, but at a much slower rate than the other proposed tests. For the test based on $\gamma = 1$ under the RKHS class (“FixedRKHS”), we observe good type-I

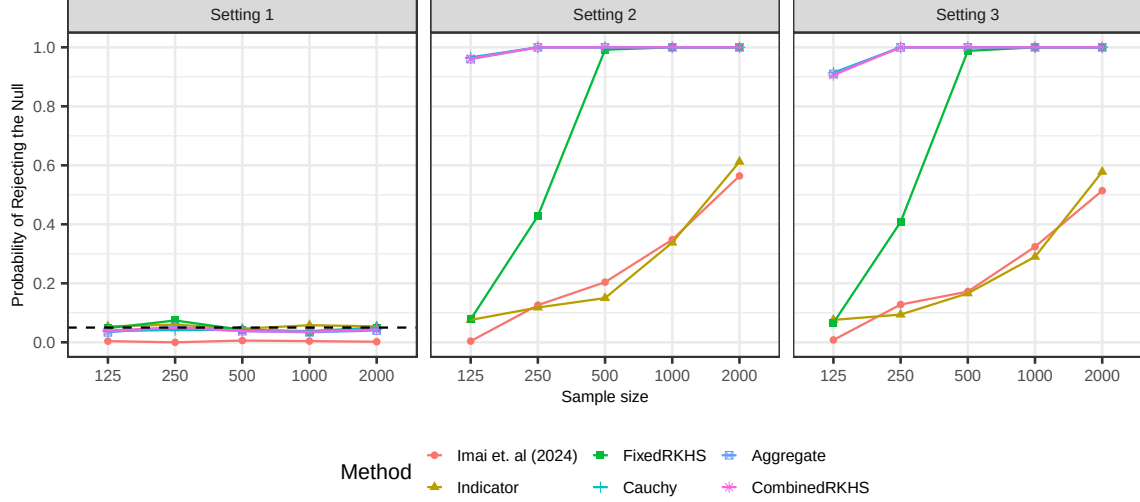


Figure 2: Empirical probability of rejection for hypotheses in Example 3 under different data generating mechanisms, at different sample sizes. Horizontal wide-dash line indicates the nominal 0.05 test size.

error control but very low power in both Settings 2 and 3. This occurs because the FixedRKHS test uses a more restrictive function class, which limits the ability of the test to detect departures from the null.

Example 2: Assessing presence of treatment effect heterogeneity

We generate our treatment assignment variable T randomly from $\{1, 0\}$ with probability 0.5 corresponding to a randomized experiment, we generate our conditioning covariate $W \sim \text{Unif}(-1, 1)$, and we generate white noise $\varepsilon \sim N(0, 0.5)$ that is independent of W . We consider three settings for the outcome distribution, corresponding to different specifications of the CATE:

- Setting 1: $Y = \beta_0 + \beta_1 W + \gamma T + \varepsilon$
- Setting 2: $Y = \beta_0 + \beta_1 W + (\gamma_0 + \gamma_1 W)T + \varepsilon$
- Setting 3: $Y = \beta_0 + \beta_1 W + (\gamma_0 + \gamma_1 \sin[W])T + \varepsilon$

Under Setting 1, there is no average treatment effect heterogeneity. Under Settings 2 and 3, respectively, there is linear and nonlinear effect heterogeneity. We estimate the nuisance functions $\pi_0(t|x)$ and $\mu_0(t, x)$ using generalized additive models (Hastie and Tibshirani, 1990) with natural cubic spline components, implemented in the R package `mgcv`. For each setting, we compare our proposed tests with the recently proposed test of Imai and Li (2024), implemented in the R package `evalITR` (Li

et al., 2023a). This method assesses treatment effect heterogeneity by testing equality of the sorted group average treatment effects (GATES) proposed by Chernozhukov et al. (2017). All tests exhibit good control of type-I error, even when the sample size is small. The approach by Imai and Li (2024) is conservative as the sample size increases in Setting 1. Under Settings 2 and 3, the Aggregate and Cauchy tests achieve high power even for $n = 125$. In these settings the low power of the test by Imai and Li (2024) may be due to the sample splitting requirement of the test.

7 Treatment effect heterogeneity in HER2-positive breast cancer

HER2-positive breast cancer is a subtype driven by over-expression of the human epidermal growth factor receptor 2 (HER2) protein, often resulting from an excess of HER2 gene copies in cancer cells. This over-expression accelerates cell growth and progression, leading to a more aggressive disease course without treatment. According to the American Cancer Society, approximately 15–20% of women with breast cancer are diagnosed with HER2-positive disease. The standard treatment for HER2-positive patients includes targeted therapies like trastuzumab, an intravenous antibody that inhibits HER2 production and has been shown to extend disease-free survival in affected women (Joensuu et al., 2006).

Additional findings by Prat et al. (2014), based on the NeoAdjuvant Herceptin (NOAH) clinical trial, suggest that HER2-positive patients classified into the HER2-enriched (HER2-E) subtype have higher 3-year pathologic complete response (pCR) rates when treated with trastuzumab and chemotherapy compared to chemotherapy alone. Classification was performed using PAM50, a diagnostic test based on the expression levels of 50 genes. However, the specific genes driving these differences in response to trastuzumab within the HER2-positive population remain unclear.

We apply our proposed test to investigate whether there is evidence against effect homogeneity with respect to gene expression biomarkers. We use data from the NOAH trial (GEO accession GSE50948), which includes 334 participants, 235 of whom were HER2-positive. The HER2-positive patients were randomized to receive either trastuzumab with neoadjuvant chemotherapy ($T = 1$) or neoadjuvant chemotherapy alone ($T = 0$). Baseline tumor gene expression levels passing quality control were available for $n = 111$ HER2-positive patients. The primary outcome is pCR within three years, defined as the absence of invasive breast cancer after treatment.

We consider nine genes which, based on the work of Prat et al. (2014), demonstrate predictive

Method	ACTR3B	BAG1	BLVRA	CCNE1	CDC20	ERBB2	FGFR4	MYC	SFRP1
Aggregate	0.210	0.034	0.177	0.556	0.647	0.447	0.990	0.917	0.849
Cauchy	0.944	0.031	0.971	0.814	0.283	0.764	0.350	0.409	0.551
Imai et al. (2024)	0.397	0.124	0.355	0.360	0.149	0.449	0.715	0.332	0.210

Table 1: p -values from the analysis based on the NOAH study. The null hypothesis is that there is a constant average treatment effect across strata of each of the nine genes.

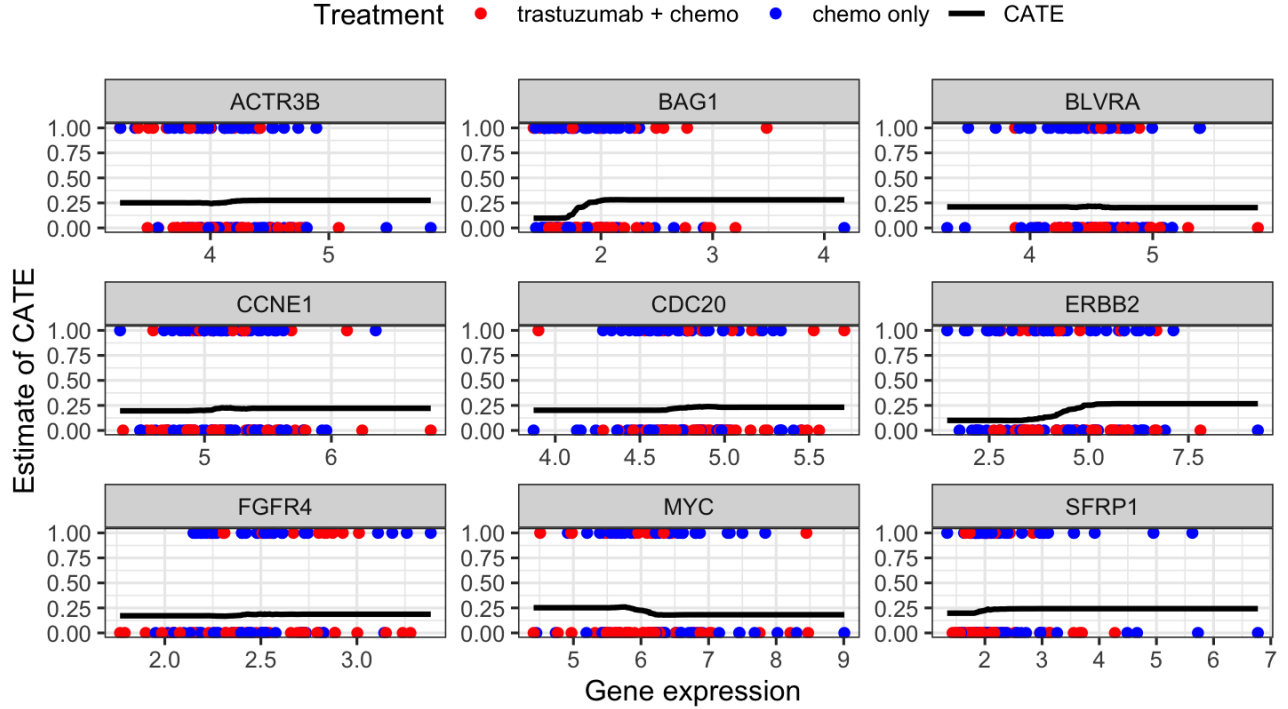


Figure 3: Estimate of CATE given each of the 9 genes using causal forest.

power comparable to the full 50-gene panel used in the PAM50 test. Table 1 presents the p -values from the proposed tests (Aggregate and Cauchy) along with those from the test developed by Imai and Li (2024). The Aggregate and Cauchy tests are implemented using the same grid of λ values as in the simulation study. Among the nine genes, both the Cauchy and Aggregate tests yield p -values below 0.05 for the BAG1 gene. In contrast, the test by Imai and Li (2024) yields its lowest p -value of 0.100 for the ERBB2 gene. All other genes exhibit large p -values across all tests.

Figure 3 illustrates the estimate of CATE given each of the nine genes. BAG1 shows strong evidence of non-constant CATE, suggesting the presence of treatment effect heterogeneity. This pattern indicates that patients with relatively low BAG1 expression may benefit less from trastuzumab compared with chemotherapy alone, whereas patients with high BAG1 expression may derive more

benefit.

8 Conclusion

We present a general nonparametric test for assessing whether the evaluation of a smooth statistical functional at a conditional distribution is constant in the conditioning variables. We develop a test that is based on representing norms as a supremum taken over a collection of finite-dimensional statistical parameters indexed by functions in a prescribed function class. Our approach ensures control of the type-I error rate for any appropriately selected smooth function class. Notably, implementing our test only requires the derivation of the first-order influence functions, and we showed that in many cases, influence functions can be readily obtained by leveraging existing results. Specifically, we established results that provide the explicit form of the EIF when the statistical functional of interest is a linear transformation of the conditional mean, or when the statistical functional does not depend on the distribution of the conditioning variables.

The choice of function class $\mathcal{H}$ can affect the power of the test. To improve the power, we proposed an aggregate test that combines test statistics obtained from a collection of different function classes. This aggregate test can often be more powerful, but the optimal choice of function class, or tuning parameter selection in terms of the RKHS remains a challenging problem in nonparametric testing settings.

Our work offers several future research directions. Firstly, our results require that the nuisance functions and their estimators are not overly complex; that is, they must belong to a Donsker class. This may preclude use of some common estimators such as random forests. One approach to relaxing this condition is constructing a cross-fitted version of our estimators ([Chernozhukov et al., 2018a](#)). Secondly, in this work, we discussed two function classes: the class of thresholded indicators and RKHS induced by the Sobolev space; however, other function classes such as functions with bounded variation norm, functions with finite VC dimension (e.g., finite-depth trees), and RKHS with other kernels, could also be used. A formal analysis of how power is affected by different choices of function classes would be an interesting topic of future research.

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